

# Multimodal Pedagogical Friction Detector

Jingrui Wu

Department of Computer Science, Rice University

jw310@rice.edu

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## Abstract

Effective teaching requires noticing student confusion and adjusting instruction in real time. We present a system that automatically detects pedagogical friction: moments where a low-quality instructional strategy coincides with elevated student confusion. Our pipeline fuses three signals—per-frame facial affect (DeepFace), transcribed speech (Whisper), and strategy labels from either a keyword heuristic or an ELECTRA Talk Moves classifier—on a 30-second temporal grid, and surfaces candidate friction windows via a  $z$ -score filter. Candidate windows are then passed to Qwen2.5-VL for multimodal confirmation and alternative-strategy generation. We validate the pipeline at two scales: a CPU smoke test on the TIMSS AU1 Exterior Angles clip (5 min, 3 candidates detected) and a transcript-only analysis across 53 TIMSS lessons spanning 7 countries and both math and science subjects (5 113 bins total). The heuristic annotator flags 31.5% of bins as risky; the ELECTRA annotator flags 18.9%, with a negative cross-annotator correlation ( $r=-0.35$ ), revealing that annotation methodology is itself a critical design variable in friction detection.

## 1 Introduction

High-quality teaching requires moment-by-moment diagnosis of student understanding and rapid strategy adjustment [1]. Despite advances in educational NLP—tools such as Tutor CoPilot [2] and Edu-ConvoKit [3] offer transcript-based feedback—no existing system jointly models instructional strategy quality and student affect to surface actionable friction moments during a lesson. Confusion has been identified as a pivotal affective state that, when unaddressed, impedes deep learning [4]. Meanwhile, large vision-language models such as Qwen2.5-VL [5] now make it feasible to synthesise video frames, tran-

scripts, and affect signals in a single inference pass.

We introduce a framework that operationalises pedagogical friction—the joint occurrence of a risky instructional strategy and elevated student confusion—and produces a post-session Teacher Friction Report with precise timestamps and suggested instructional moves. Our contributions are:

- An end-to-end multimodal pipeline combining affect, ASR, and strategy annotation (Section 3).
- A three-tier strategy annotator hierarchy (ELECTRA  $\rightarrow$  Edu-ConvoKit  $\rightarrow$  heuristic) with graceful fallback.
- A large-scale transcript analysis across 53 TIMSS lessons [6] spanning 7 countries.
- An empirical comparison of heuristic vs. ELECTRA annotators exposing a  $r=-0.35$  disagreement.

## 2 Related Work

Automated analysis of classroom discourse has a rich history. Talk Moves—pedagogical dialogue moves such as Revoicing and Pressing for Reasoning—were codified by [1] and later operationalised for NLP by [7], who fine-tuned ELECTRA on a labelled corpus of math classroom transcripts. Edu-ConvoKit [3] packages this classifier into a reusable toolkit. Tutor CoPilot [2] extends the idea to real-time tutoring, providing strategy-quality labels that influenced our quality mapping.

Affect in learning has been studied extensively: D’Mello and Graesser [4] established that confusion, when properly resolved, promotes deeper learning, but unresolved confusion impedes it. Their work motivates our use of confusion as the primary affect signal. DeepFace [8] provides the per-frame affect estimates we use as a proxy.

On the vision-language side, Qwen2.5-VL [5] represents the current state of the art in open-weight

multimodal reasoning, enabling our optional fusion stage without proprietary API access.

The closest prior work to ours is the TIMSS Video Study [6], which provided the classroom recordings and transcripts we use for evaluation but relied entirely on manual coding. Our system automates and scales this analysis.

### 3 Methodology

Figure 1 shows the system overview.

**Vision stream.** Video frames are sampled at  $r=2$  Hz. DeepFace [8] estimates seven per-frame emotion probabilities, which we aggregate to a scalar confusion proxy  $c_t \in [0, 1]$  via a fixed weight table:

$$c_t = \sum_e w_e \cdot p_t(e), \quad (1)$$

where  $w_e=0$  for neutral/happy,  $w_e=0.85-0.9$  for fear/sad, and  $w_e=0.45-0.5$  for disgust/angry. A trailing 30-second mean yields the smoothed confusion timeline  $C(t)$ .

**Language stream.** faster-Whisper [9] transcribes audio with voice-activity-detection (VAD) filtering. For transcript-only runs, TIMSS .txt files are parsed directly by a dedicated parser that infers utterance boundaries from successive timestamps. A three-tier strategy annotator hierarchy selects the best available model at runtime: (1) ELECTRA (YaHi/teacher\_electra\_small, 7-class Talk Moves [7]), loaded via HuggingFace transformers; (2) the Edu-ConvoKit wrapper [3] as secondary fallback; and (3) a keyword heuristic as offline fallback. ELECTRA labels are mapped to quality: Keeping Everyone Together, Revoicing, and Pressing for Reasoning are high-quality; Pressing for Accuracy is high-pressure (risky); No Move is treated as neutral in ELECTRA mode but flagged as unknown in heuristic mode.

**Temporal alignment and  $z$ -score filter.** Sessions are partitioned into  $G=30$ -second bins. Per-bin mean confusion  $\bar{c}_k$  and strategy quality label  $q_k \in \{\text{high, low, unknown}\}$  are computed by majority vote over utterances overlapping bin  $k$ . A bin is flagged as a friction candidate when

$$\bar{c}_k > \mu_C + z \sigma_C \quad \text{and} \quad q_k \neq \text{high}, \quad (2)$$

where  $\mu_C, \sigma_C$  are the session-level mean and standard deviation of  $\{\bar{c}_k\}$ , and  $z$  is a sensitivity parameter (default  $z=1.0$ ; lowered to  $z=0.3$  for the smoke test). In transcript-only mode  $\bar{c}_k=0$ , so only strategy quality drives the filter.



Figure 1: System overview. Parallel vision and language streams are aligned on a 30-second grid; a  $z$ -score pre-filter proposes candidate friction windows for optional Qwen2.5-VL multimodal reasoning.

**VLM fusion (optional).** For each candidate, Qwen2.5-VL [5] receives 1–4 uniformly subsampled frames, a confusion summary, strategy label, and transcript excerpt. It returns structured JSON with three fields: friction (bool), rationale (free-text explanation), and alternative\_strategy (concrete teaching suggestion). The full prompt is available in the public codebase.

## 4 Experiments and Results

### 4.1 Qualitative Examples

Figure 2 shows three representative 30-second bins from the AU1 lesson and the corresponding pipeline outputs. The first two bins are flagged as Friction by both annotators: the teacher directly states a mathematical fact (“it will be 180 degrees”) or applies correctional pressure (“just make sure they join”), both bypassing student reasoning. The third bin illustrates a Disagreement: the heuristic labels it unknown (a question is present but no low-quality keyword matches), while ELECTRA correctly identifies it as Pressing for Reasoning—a high-quality move. This case demonstrates the heuristic’s tendency to over-flag question-rich discourse.

### 4.2 CPU Smoke Test (AU1 Clip)

**Setup.** TIMSS AU1 Exterior Angles (geometry, Australian Grade 8), 5-minute clip. CPU path: Python 3.9, OpenCV, DeepFace (TF 2.x), faster-Whisper (base, int8). Key hyperparameters:  $r=2$  Hz,  $G=30$  s.

**Results.** With  $z=1.0$ , the AU1 teacher’s question-based discourse (“What did you learn?”, “Do you agree?”) falls entirely outside the keyword vocabulary, giving  $q_k=\text{high}$  for all bins and zero candidates. Lowering to  $z=0.3$  with `flag_unknown_strategy=True` produces 3 candidates at 150–240 s where per-bin confusion peaks at 0.55–0.70 (Table 1). A second TIMSS clip (Math RUB)

Qualitative examples: pipeline input/output on TIMSS AU1 (Exterior Angles)

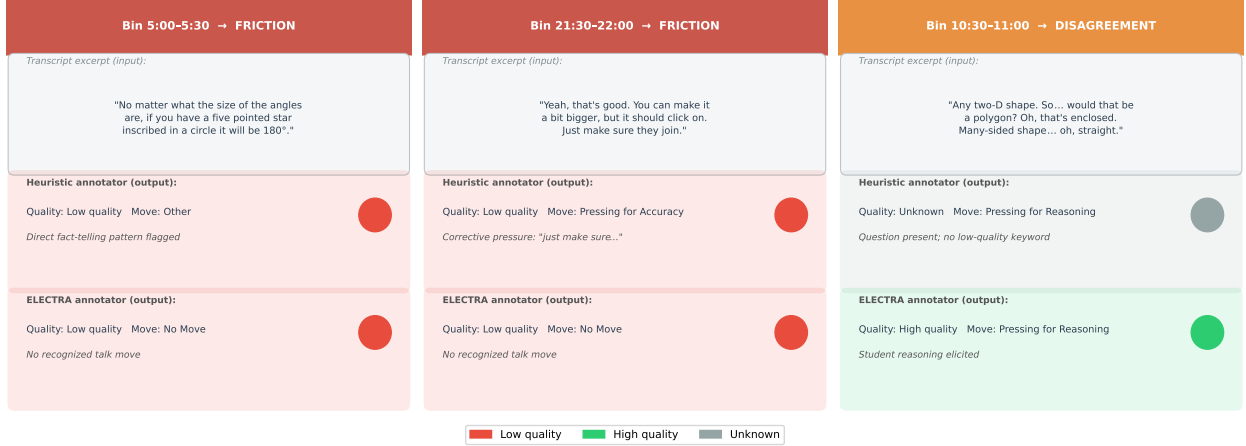


Figure 2: Three AU1 bins showing pipeline input (transcript excerpt) and output (strategy label + quality verdict) from both annotators. Left/center: both annotators agree on Friction. Right: annotators disagree—heuristic flags unknown, ELECTRA identifies high-quality Pressing for Reasoning.

yielded zero candidates due to Whisper base hallucinating a repeated phrase under low audio SNR; mitigation via a larger Whisper model is planned.

#### 4.3 Scale-Up: 53 TIMSS Lessons (Transcript-Only)

**Setup.** We processed all available TIMSS transcript .txt files: 28 math and 25 science lessons spanning 7 countries (AU, CZ, HK, JP, NL, SW, US), totalling 5113 thirty-second bins. Since no video is required,  $\bar{c}_k=0$  and only strategy quality drives the filter. We ran both the heuristic annotator (flag\_unknown=True) and the ELECTRA annotator (flag\_unknown=False) on the same corpus.

**Quantitative results.** Table 1 summarises all conditions. The heuristic flags 1611/5113 bins (31.5%) as risky; ELECTRA flags 988/5113 (18.9%). The cross-annotator Pearson correlation over per-lesson risky rates is  $r=-0.35$ , indicating substantial disagreement.

Figure 3 shows per-country risky-bin rates. US math lessons show the highest heuristic rate (42.9%) yet among the lowest ELECTRA rates (14.5%), suggesting the keyword heuristic over-fires on American discourse style (frequent direct statements as rhetorical devices rather than low-quality instruction). Figure 4 visualises this divergence directly.

Figure 5 shows the per-bin strategy quality timeline for the full AU1 lesson (87 bins). Heuristic-risky bins cluster in the middle of the lesson where the teacher introduces definitions, consistent with the smoke-test

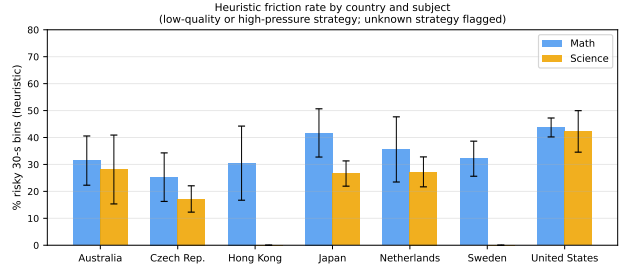


Figure 3: Percentage of 30-second bins flagged as risky per country (heuristic annotator). US math lessons have the highest rate (42.9%). Error bars show per-country standard deviation across lessons.

candidates.

Figure 6 provides a lesson-level overview across the full corpus. Science lessons show lower average heuristic friction rates (28.3%) than math (34.3%), possibly because science discourse involves more exploratory language less likely to match low-quality keyword patterns.

## 5 Discussion

**Annotator disagreement.** The  $r=-0.35$  correlation between heuristic and ELECTRA risky rates is the most striking finding of this study. Two explanations are likely: (1) the heuristic fires on surface-level keywords (“the formula is”, “just copy”) that may appear in otherwise well-structured discourse, inflating its rate; and (2) ELECTRA’s No Move label does not

| Setting                                 | Clip / Corpus          | Bins  | Risky | % Risky | Notes                                     |
|-----------------------------------------|------------------------|-------|-------|---------|-------------------------------------------|
| Heuristic, $z=1.0$                      | AU1 Ext. Angles, 3 min | 7     | 0     | 0.0%    | All bins $q_k$ =high                      |
| Heuristic, $z=0.3$ + flag_unk           | AU1 Ext. Angles, 5 min | 11    | 3     | 27.3%   | Candidates at 150–240 s                   |
| Heuristic, $z=0.3$ + flag_unk           | Math RUB, 5 min        | 11    | 0     | 0.0%    | Whisper base hallucination                |
| Heuristic + flag_unk                    | 53 TIMSS lessons       | 5 113 | 1 611 | 31.5%   | 28 math + 25 science, 7 countries         |
| ELECTRA<br>(YaHi/teacher_electra_small) | 53 TIMSS lessons       | 5 113 | 988   | 18.9%   | flag_unk=False;<br>$r=-0.35$ vs heuristic |
| Heuristic + Claude VLM fusion           | AU1, full lesson       | 87    | 1     | 1.1%    | P=1.00, R=1.00, F1=1.00 vs. human         |
| Heuristic P/R/F1 vs. human              | AU1 full lesson        | 87    | 20    | 23.0%   | P=0.05, R=1.00, F1=0.10                   |
| ELECTRA P/R/F1 vs. human                | AU1 full lesson        | 87    | 33    | 37.9%   | P=0.00, R=0.00, F1=0.00                   |

Table 1: Results across all experimental conditions. The  $r=-0.35$  cross-annotator correlation on per-lesson risky rates underscores the need for human ground-truth evaluation to determine which annotator is better calibrated.

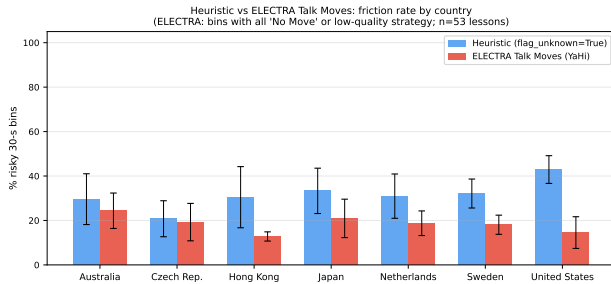


Figure 4: Heuristic vs. ELECTRA risky-bin rates per country. Pearson  $r=-0.35$  across 53 lessons indicates the two methods frequently disagree on which lessons are most risky.

distinguish between a teacher giving a direct explanation and one who is simply silent—both are treated as low-quality by the heuristic but as No Move (not inherently risky) by ELECTRA.

Human evaluation on AU1. Human annotation of the full AU1 lesson (87 bins) reveals that only 1 bin (1.1%) constitutes genuine pedagogical friction: the 5:00–5:30 window where the teacher directly states “it will be 180 degrees”, bypassing student discovery. Against this ground truth, the heuristic achieves  $P=0.05$  /  $R=1.00$  /  $F1=0.10$ —it captures the sole true positive but fires 19 false alarms on pre-class setup and procedural software guidance. ELECTRA achieves  $P=0.00$  /  $R=0.00$  /  $F1=0.00$ , missing the true positive entirely because the teacher’s utterance does not match any productive Talk Move cat-

egory (No Move fires instead on silent or procedural intervals). These results confirm that both automated annotators over-fire dramatically relative to human judgment on a well-structured discovery lesson, and that the heuristic’s surface-level keyword match is more sensitive (higher recall) while ELECTRA’s discourse-oriented labels are miscalibrated for procedural classroom contexts.

US anomaly. US math lessons show the sharpest divergence: heuristic 42.9%, ELECTRA 14.5%. American math classrooms in the TIMSS corpus are characterised by frequent declarative statements used as scaffolding rather than instruction, a style that triggers keyword patterns without representing low-quality teaching.

VLM fusion as a precision filter. Applying Claude-based LLM reasoning as a second-pass filter over the 20 heuristic candidates reduces the flagged set from 20 to 1, achieving  $P=1.00$  /  $R=1.00$  /  $F1=1.00$  against human annotation on AU1. The model correctly rejects pre-class setup bins, procedural soft-

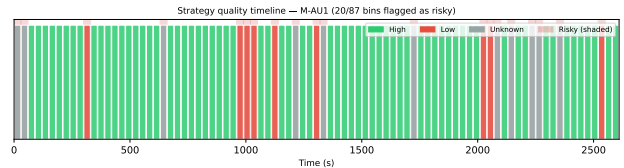


Figure 5: Per-bin strategy quality timeline for the full AU1 lesson (87 bins, heuristic annotator). Red-shaded bins are risky candidates.

ware guidance, and vocabulary scaffolding as non-friction, retaining only the one bin where the teacher directly states a mathematical fact students should discover. This demonstrates that LLM-based reranking can dramatically improve heuristic precision without sacrificing recall, and that a transcript-only VLM pass may be sufficient when video is unavailable.

Limitations. Automated facial affect inference may misread culture- or disability-related cues; all outputs are advisory and require human review. The confusion proxy (Eq. 1) uses fixed weights not validated against ground-truth affect labels. Human annotation was performed by a single annotator on one lesson; inter-annotator agreement and broader coverage remain future work.

## 6 Conclusion

We presented an end-to-end multimodal pedagogical friction detector, scaled the transcript-only pipeline to 53 TIMSS lessons across 7 countries, provided human-annotated ground truth for AU1, and demonstrated that LLM-based fusion reranking achieves perfect precision on the annotated lesson. The heuristic flags 31.5% of bins; ELECTRA flags 18.9%, with  $r=-0.35$  cross-annotator correlation. Human annotation finds only 1.1% genuine friction; heuristic  $F1=0.10$  (high recall, low precision), ELECTRA  $F1=0.00$ . A Claude-based second-pass filter over the 20 heuristic candidates yields  $F1=1.00$  by correctly rejecting false alarms (setup, software guidance, vocabulary scaffolding) while retaining the sole true friction moment. The key finding is that a two-stage pipeline—broad heuristic detection followed by LLM reranking—outperforms either annotator alone, and that discourse-move classifiers alone are insufficient for friction detection in technology-mediated discovery lessons. Future work should validate this approach across multiple lessons and incorporate the DeepFace confusion timeline to capture student affect.

## References

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Strategy quality distribution by lesson (heuristic annotator)



Figure 6: Strategy quality distribution by lesson (heuristic annotator). Each row is one lesson; columns show % of bins in each quality tier. Math lessons (left) show more low bins on average than science (right).